Yuval Ardenbaum

Tokens are all you need

Re-draft, 2026-10-01. Not for circulation.

Abstract

Accelerated compute is a non-storable commodity with a negative cost of carry. Whoever holds inventory of GPU hours (contracts or spot) pays a storage cost to do so, due to the provider's overhead of keeping the hardware reserved. This cost can't be hedged away, because there is no cash-and-carry arbitrage in an asset that expires hourly.

In spite of increasing scarcity of access and sophisticated fleet scheduling and optimization middleware, accelerator fleets are woefully underutilized right now due to the variance & inefficiency of transactions in the industry. Price discovery is fragmented & vibes-based at best & venture dollars are burning without productively contributing to superintelligence.

With inference fast becoming the dominant productive use of compute, and exponential growth of the customer base for open weight models predicted (and actualizing), it's clear that inference is the most liquid market for price discovery in the underlying.

I propose a preliminary design for a platform inspired partly by the architecture of decentralized money markets and AMMs. This venue converts wasted compute into revenue, by matching holders of idle GPU hours with pools of inference providers (aggregated per model & filtered by attributes like quantization), then enabling those providers to serve verified tokens to customers. This subsidizes carry costs, supports a truly efficient market, enables builders' ability to serve products to their customers at predictable cost, and allows the market makers supporting pricing efficiency to converge on delta-neutrality.

Three conditions must hold before such a venue can function as intended:

1. With regard to the transaction between inference providers & holders of compute - the traded object has to be fungible, which it is not today. The resource itself consists of heterogeneous SKUs: GPU-hour offtake agreements are struck case-by-case, each with its own tenor, location, chip and covenants, so no two are the same instrument. This makes grouping of contracts difficult, and fragments price discovery. Fungibility is fundamental.

2. That said, even a standardised hour does not guarantee a standardised output, because providers can, and do, serve degraded models on aggregators and elsewhere to capture spread (see Section 4).

In the presence of fraud, the unit economic forces that drive efficient supply & demand curves are diluted in ways that adversely affect all honest participants. Pricing fairly requires verification.

3. The clearing mechanism has to respect the physics of the underlying. Model load times measured in tens of seconds in most cases, locality constraints on data, and the non-storability of the hour influence how inference providers consume compute, and how the ultimate inference customer consumes tokens. Matching & clearing should derive in part from solutions already studied deeply in wholesale electricty markets, such as constrained dispatch (while acknowledging the greater problem of solving for heterogeneity in compute SKUs).

Such a venue, at scale, would produce records of performance & buying power, and would enable issuance of fungible contracts (block trades) against the venue itself as a counterparty. Therefore, it would provide a basis for offtake-financiers to lend against. Lenders in this category are already sitting on billions of dollars in deployable capital, and need only underwrite the contract and counterparty (the platform in this case), so credit is a module the venue may expose (via third parties) once initial liquidity has been bootstrapped.

What the platform enables is sale of the compute resource (spot or forward contracts for enterprise use cases) to inference providers, and subsequent sale of verified tokens to consumers, resulting in efficient price discovery at a scale that doesn't currently exist.

As the world trends toward democratization of superintelligence, builders should be enabled to focus on shipping product to their customers, and purely financial participants should be able to monetize compute resources without needing to productize an inference stack. This is why a productized compute ops stack must exist, with abstractions to accommodate both simple & more sophisticated demand.

---

1. Introduction

1.1 TIdr;

Compute is a commodity that expires every hour, and there are many market participants allowing GPU-hours to evaporate unused. There is currently no easy way to re-monetize these hours. The solution I'd propose is a two-leg venue that puts those hours to work serving inference, first by facilitating exchange between GPU-hour holders (forward & spot), and then by facilitating the sale of tokens.

Two things have to be true first. The hour has to be a standard product, which means pinning the hardware, the region and the model to a homogeneity class, and then proving the model served is the model advertised. The market should also clear the way electricity does, by committing capacity ahead of time and dispatching under physical constraints, because a GPU hour cannot be bought the instant it is needed (even in spot markets, load times create latency to first token).

Part 1 of this paper works through what that implies: who the participants are, why the yield on idle compute is currently too correlated, how verification is built, and how a two-leg exchange architecture might draw inspiration from clAMMs, decentralized money markets & mining pools. Part 2 will cover how clearing works, where credit plugs in, and how to solve the liquidity cold start problem (hint: HLP).

1.2 Utilisation is the only lever

An inference business has a cost structure dominated by the capital cost of accelerators, which is fixed at the point of acquisition and depreciates on a schedule (albeit an increasingly unpredictable one; discussed later) set by the next hardware generation rather than by usage. Against a fixed cost base, the only variable that moves unit economics is utilisation. Every other lever, including model selection, serving stack optimisation, and pricing, is ultimately a utilisation strategy.

Inference clouds nonetheless compete on differentiation. I take the view that in many cases (not all), this is an artifact of the funding model rather than a strategic necessity. Venture capital underwrites GPU capex against an expectation of venture returns, and venture returns require a defensible product story & brand equity. Differentiation is therefore a constraint imposed from the capital structure downward; the unit economics don't justify it, except in cases of meaningful, proprietary innovation at the hardware or software/orchestration layer.

Relax that constraint and the objective function changes. A business optimising for return on deployed capital rather than for enterprise value has no reason to fear commoditisation. If the asset is commoditised and you are the venue on which it trades, commoditisation is the thesis. Likewise if you are in the business of commodity delivery, your economics become more predictable, and thus, more financeable.

1.3 Aggregation provides a marketplace, not market structure

The incumbent layer between inference demand and inference supply consists of routing aggregators. They present a unified API, maintain a list of providers, and route on posted price and observed latency.

This is a price comparison and attribute optimization service, not a market. It has no clearing, no standardised contract, no settlement guarantee, no margin, and, most consequentially, no mechanism or incentive to police quality. These are payment businesses, and are incentivized to allow whatever behaviour results in the greatest volume.

A provider advertising a model at a given precision can serve a more aggressively quantised variant, a distilled substitute, or a shorter effective context, and capture the difference between the advertised cost basis and the real one. Detection is statistically awkward because model outputs are stochastic, and no aggregator I'm aware of runs continuous adversarial verification. The result is a market in which the cheapest quoted price is often the least trustworthy; a classic example of a market for lemons.

The ability to serve a worse model than advertised directly interferes with pricing efficiency

across the inference market at large, and therefore verification as a native feature is essential to standardizing the underlying asset and an indispensable part of this venue's design.

---

2. Market participants

Inference has overtaken training as the dominant use of GPU hours - various estimates put inference at roughly two-thirds of all compute consumed in 2026.

The demand profiles are quite different - training demand materializes in large, lumpy, negotiated blocks from a few buyers at a time. Comparatively, inference demand arrives continuously, in small units, from many buyers, priced per token & mostly on public venues (albeit not very transparently).

Serving tokens has a price at every minute of the day, making it the most liquid candidate price discovery mechanism for compute. Every bid is a live datapoint. And there are far more continuous, publicly exposed bids for inference than there are for underlying compute contracts.

That said, inference is a nuanced and highly competitive business in its own right across several vectors:

The serving stack: model loading, batching, caching, quantisation - the software stack that converts compute hours into tokens.

The product: API, accounts, devex, rate limits, uptime, support.

Distribution: growth, sales & customer support motions that influence customer decision-making between one provider and another.

While verticalized inference providers, neoclouds and the like are natural holders of idle GPU hours, there are many compute consumers who don't have this capability built out, such as startups, research labs, academic institutions, market makers & prop firms.

Even among neoclouds, fleet utilization rates are not always optimal due to the 'take-or-pay' nature of most modern offtake agreements[1]. Standing up inference as a new business line is expensive and competitive, and most organisations are not structurally suited to it.

Ultimately, inference providers need competitively priced compute, and owners of compute resources often don't have the infrastructure to consume this resource profitably or in full. A venue matching the two parties inverts the current supply-side pricing mechanics that derive price from the rental costs of the underlying hardware, to a demand-side pricing mechanic that

[1] Note that this disregards contracts with hyperscalers and the top bracket of frontier labs that have the purchasing power to mandate exclusivity. The long tail of non-exclusive demand is big enough to sustain this product, and growing exponentially.

drives pricing power from offtakers by aggregating competition. You can think about this like Robinhood - retail pricing power was sufficient but too fragmented to move markets in prior years. However, during the Gamestop squeeze, we saw the power of aggregating the retail bid on one venue with a shared mythos, culture and incentive structure. Similarly, unifying incentive coordination under one roof, with a shared philosophy of price reduction & democratized intelligence aggregates the bid for cheaper compute resources.

There are various profiles of inference providers that might serve as takers on a venue like this, and they each drive healthy pricing competition in unique ways:

Operator type	**Owns**	**Elasticity to capacity price**	**Credit quality (by traditional credit-worthiness)**
Vertically integrated	Hardware (sometimes proprietary) and serving stack (always proprietary)	Low, already supplied	Strongest, has a balance sheet
Software optimised, leased compute	Serving stack only	High, capacity is the input cost	Moderate, asset light
Opportunistic ‘searchers’ (term borrowed from blockchain MEV) focused on innovation at the runtime layer	Routing logic only	Extreme, exists to arbitrage capacity price	Weakest, limited balance sheet

The edge each provider type competes on is also predictive of their need - verticalized providers seek only to optimize their fleet as close to maximum utilization as possible. Meanwhile, providers focused on their serving stack typically compete on other attributes, like throughput or TTFT.

An emergent class of provider focuses on edge in the inference runtime, using kernel optimization or specialized decoders to create viable unit economics on leased spot hardware with no preferential access.

Together, they create a demand liquidity ecosystem (and maximum competition when pooled) for all forms of GPU hours - spot & contracts with various tenors, dating, locations and other attributes.

3. The economics of a non-storable commodity

3.1 There is no cash and carry

In a storable commodity market, the forward price cannot exceed spot plus the cost of storage and financing, because any wider basis is closed by arbitrage - buying spot, storing, and delivering into the forward. That relationship anchors the curve & introduces predictable pricing mechanics.

Compute has no such anchor - a GPU hour contracted for a specific timeframe evaporates if unused. A good comparison is wholesale electricity, where the absence of storage results in forward curves that decouple from spot, volatility an order of magnitude above storable commodities, and squeezes that can't be defended by building inventory. Any market design built on implicit assumptions from storable commodity microstructure will mis-specify the risk, and fail to facilitate the most efficient pricing possible.

3.2 The refinery hypothesis - the value of producing a finished good from raw materials

Consider a market maker holding inventory on a physically-settled compute exchange, like Ornn (an example of a well-built, prototypical physical delivery exchange). Let the acquisition cost of a GPU hour be *c*, and let the holding cost of an unsold hour be *h*. Let *h* be $3.25.

Ornn's reserved bookings are paid upfront in full at checkout, and 'spend' (debit) against the reservation begins at T0 of the contract, not at VM deployment. There is no way to pause this spend or refund it from T0.

A relist feature that enables holders to post their reservation back into an orderbook exists, though this assumes sufficient spot liquidity and just-in-time demand to consume the relisted resource bootstrapped concurrently, which is rarely the case.

As a result of liquidity cold-start, like crypto markets, many exchanges have and will continue to engage with incentivized market makers who are paid in some form not to cross the spread. As such, they are prohibited from selling back into the book.

In other markets, these maker positions can be delta-hedged on a future (perp or otherwise) because the instruments traded are intangible, and physical settlement just means delivery of the underlying asset to a wallet or account, rather than delivery of a useful resource. The delta hedge makes providing liquidity economically viable, and therefore bootstrapping new venues has become *relatively capital efficiency.

How would a market participant like the makers above achieve delta-neutrality or otherwise subsidize the cost of a GPU-hour position?

The auxiliary cash-settled futures on rental rates deployed by companies like Ornn, Silicon Data

& the like, do nothing for *h*, because *h* is the holding cost of a resource that expires (whether used or not), and expiration of a non-storable can't be hedged this way. Cash-settled futures also don't hedge out exposure to *c*, the cost of the hour. Instead, these contracts reduce the variance of c over a tenor. This is beneficial for making a position financeable via predictable costs, but only dampens volatility, not the price itself (except for reduction in cost of capital).

A hedge leaves *c* where it is and substitutes the forward price *F* for the realised spot price: a holder short futures at *F* who later buys at spot *S* has an effective cost of $S - (S - F) = F$. The exposure is transformed/normalized for the sake of forecasting, rather than reduced.

That substitution is not neutral, because *F* is not the expected spot price. The forward price embeds a risk premium set by the balance of hedging pressure between producers seeking to protect output prices and consumers seeking to protect input costs.

That said, while the benefit of dampening volatility in *c* is clear, there is a structural issue with this hedge: Cash settlement supplies money, not silicon.

A hedge pays out dollars when prices rise; it does not deliver capacity. In a scarcity event the correlation between "the price spiked" and "I could not source the physical" approaches certainty, so the instrument settles when it is least useful to a party who needed the compute rather than the cash. Cash is of less use to a startup that needs to serve tokens to its users to maintain uptime & reputation for growth. And it's abundantly clear the compute market can move out from underneath such a startup if they're scrambling to acquire GPU-hours urgently with the cash they received from settlement of their futures position.

Index basis here is also much wider and more warrant to drift, than in mature commodities. A cash-settled contract settles against an index; the Ornn OCPI is an average of transacted rentals across the market, while any given holder's *c* is an invoice for a specific SKU, venue, region and date. Likewise, Silicon Data's index covers its own incomplete base of transaction data. Until the dataset becomes large & pervasive enough for these indices & all *c* to converge, the hedging opportunity is imperfect at best. Further, the index composition turns over with the hardware generation, making long-dated hedges impossible.

Margin adds another consideration that is about liquidity rather than price. Variation margin requires posting cash today, while the offsetting physical benefit accrues across the delivery period, subject to the unit economics of the business. The timing mismatch has bankrupted hedgers across other asset classes, whose underlying thesis was correct.

Now let the market maker 'lend' or allocate the idle hour to an operator who serves tokens on it. Let *T* be throughput in tokens per hour, *p* the realised price per token, *u* the fraction of the hour utilised, and *m* the marginal cost of serving, which is power, egress, and operations. The maker's net carry becomes:

$$carry = h - (u \cdot T \cdot p - m)$$

When net inference demand exceeds *h*, inventory flips from negative to positive carry. A negative-carry asset cannot support a term structure, because nobody will hold it across time without compensation, and so there is nobody to take the other side of a forward. A

positive-carry asset can. Inventory becomes financeable, market makers can hold books rather than flatten them intraday, and a credit and derivatives market become possible on top. This condition is possible on existing physical settlement exchanges, but requires the GPU-hour holder to own a serving stack and/or have a fully optimized fleet in the case of hardware owners.

3.3 Efficient, competition-enforced pricing mechanisms exist in a microcosm

Jane Street notoriously allocates their internal accelerator fleets through a bidding system that encourages teams to assign a notional USD value to the work they're doing. This has resulted in an internally-conceived spot pricing mechanism for GPU-hours inside the firm.

There is no strict auditing or top down enforcement of policy, but all workloads are transparent, encouraging socially-enforced accountability for efficient usage. Though we can assume the level of adversarial behaviour is less between teams intra-organization than between competing entities, given broader shared incentives, this is a reasonable analog to open market competition. It points to existing efforts applied by organizations concerned with capital efficiency.

---

4. Fungibility requires verification

4.1 The degraded model problem

An exchange requires a contract that means the same thing to both parties. Inference doesn't have one.

A provider quoting a model at a named precision has a perverse incentive to serve something cheaper: a more aggressive quantisation, a distilled variant, a truncated context window, or degraded output quality from speculative decoding pushed to the extreme. The buyer receives plausible tokens, and provider reputation & support only suffers when the margin of difference becomes blatant.

Verification is hard because the output distribution is stochastic, so a single response carries almost no evidence, and the degradation may be small enough to be invisible in ordinary use while being large enough to matter on average.

The consequence is the unraveling of a market, described by economist Akerlof's market for lemons thesis. Honest providers can't match the quoted price of degraded ones. Buyers, unable to distinguish, price to the average. Honest providers exit or degrade. The observable symptom is that aggregator prices converge downward while quality dispersion widens, and buyers who

care about quality withdraw to direct relationships with named providers, which is where enterprise inference demand mostly sits today.

This is documented - Cai et al. (2025) [arXiv:2504.04715] frame covert substitution of quantised or smaller models as the central trust problem in commercial LLM APIs and evaluate detection against it directly.

Lin et al. (2026) [arXiv:2604.21083] ran a black-box audit of LLM API gateways and found model misrepresentation, substantial performance variability under nominally identical model selections, and billing discrepancies.

A NeurIPS 2024 from Dutta et al. [arXiv:2407.09141] result surfaces why this is more insidious than it seems: compressed models flip individual answers in both directions and degrade markedly on free-form generation while aggregate accuracy holds, so summary performance statistics benchmarked by buyers mostly hold.

With current exchange paradigms, price doesn't track precision, so the price signal can be actively misleading. For meta-llama/llama-3.3-70b-instruct, originally released in bf16, the most expensive endpoint has been observed serving fp8, a cheaper one bf16, and the cheapest fp8 again. Aggregators expose a quantisation attribute, but disclosure is both optional & unaudited.

In one study, changing only the inference backend moved a benchmark score by up to 16.6% [arXiv:2605.19537].

In another case, a NeurIPS 2025 paper reported that some reasoning models produce illegible chains of thought [arXiv:2510.27338]. When the experiment was re-run with a different provider serving the same model, the illegibility score fell from 4.30 to 2.30 (nostalgebraist, LessWrong 2026). The author accepted that his results "were contaminated by bad inference setups" (nostalgebraist, LessWrong 2026).

Verification is a necessary condition for the economic drivers of pricing efficiency to emerge, in much the same way that crude oil has benchmarks like Brent to achieve efficient price discovery (Brent is a framework that accounts for quality differentials in crude oil, just like model quantization in this analogy).

4.2 Verification mechanisms

There are many ways to skin this cat, though some require a meaningful shift in incentive structures.

Challenge requests

Inject canary prompts indistinguishable from live traffic and score responses against a reference deployment. Cheap to operate, and difficult for the operator to selectively cheat. That said, the literature's verdict is that statistical tests on text outputs are query-intensive and only successfully identify gross substitution, which is a minority of cases relative to undisclosed quantization.

Distributional fingerprinting

Compare returned token log-probabilities against a reference deployment of the claimed weights. Problematically, log-probability methods are defeated by inference nondeterminism in production environments, where batching, kernel selection and scheduling differences across providers make exact comparison unreliable - even honest actors serving the correct weights. Rank-based variants exist [arXiv:2506.06975] and are worth considering, but not the most promising candidate solution.

Weight attestation. Use confidential computing on current-generation accelerators to produce a signed measurement of the loaded weights and the serving configuration. This is the only mechanism that is dispositive rather than statistical. Cai et al. propose TEEs as the practical solution, giving provable cryptographic guarantees of model integrity at what they characterise as modest performance overhead. Realistic implementation would require a great incentive - either by unlocking greater revenue for compute providers or by penalizing their revenue.

Bonding and slashing. Require operators to post a bond exceeding the expected gain from degradation over the expected detection window, and slash on a confirmed finding.

The bond's economic function may be funded from a rolling reserve on the operator's earnings rather than from posted capital, so the incentive structure materializes without capex from the provider. It is not a detection method, but it increases the economic incentive for more useful detection methods arising.

Cai et al. tried output-based statistical tests and log-probability methods against documented dishonest behaviors, including quantisation, randomised substitution and benchmark evasion, and concluded that software-only methods are fundamentally unreliable.